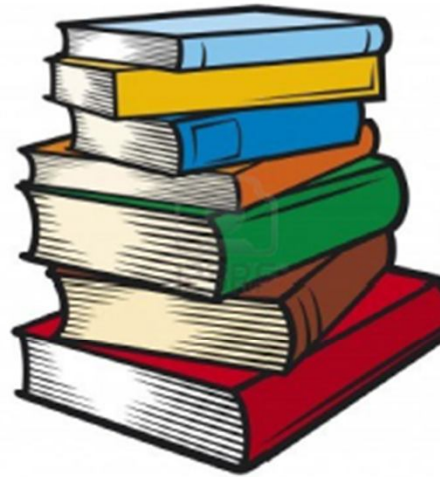


EE204 Semiconductor Physics and Devices



Logistics



Books



**Syllabus &
Grading**

Dr. Shuming Chen

Department of Electrical and Electronic Engineering, SUSTech

Instructor

Instructor: Dr. Shuming Chen (陈树明)

chen.sm@sustech.edu.cn

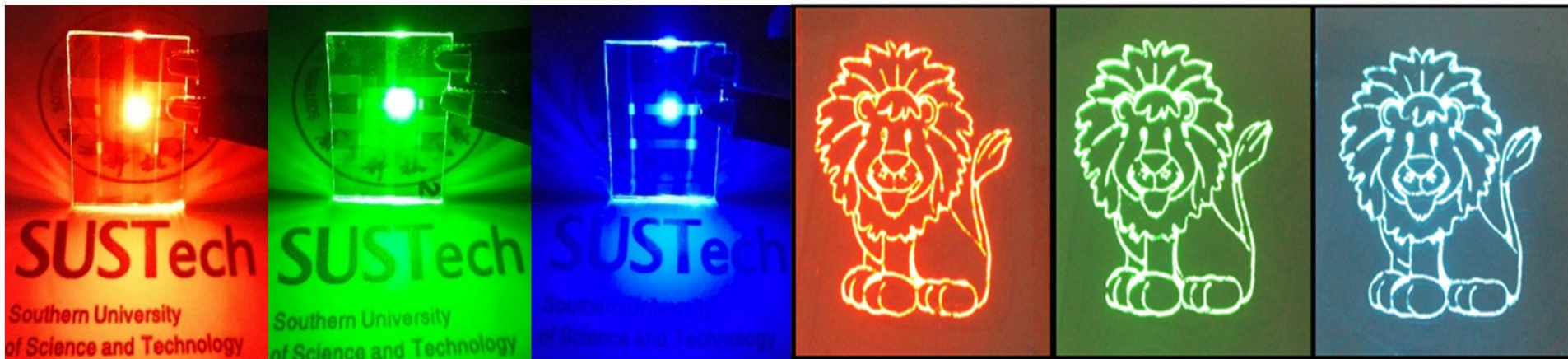
Rm 330, South Tower, Engineering Building, SUSTech

Tel. (o): 8801 8522

Office hour, Tuesday 2:00-4:00 PM

TA: 郑东泽, Rm 338, South Tower, Engineering Building

Lab: LinLin Xu (徐琳琳) xull@sustech.edu.cn



Logistics

Schedule:

❑ Tuesday, 10:20-12:10 PM, Rm 325,
#1 teaching building

❑ Lecture notes, reading materials,
problem sets and etc will be
uploaded to:

QQ group: 187161534

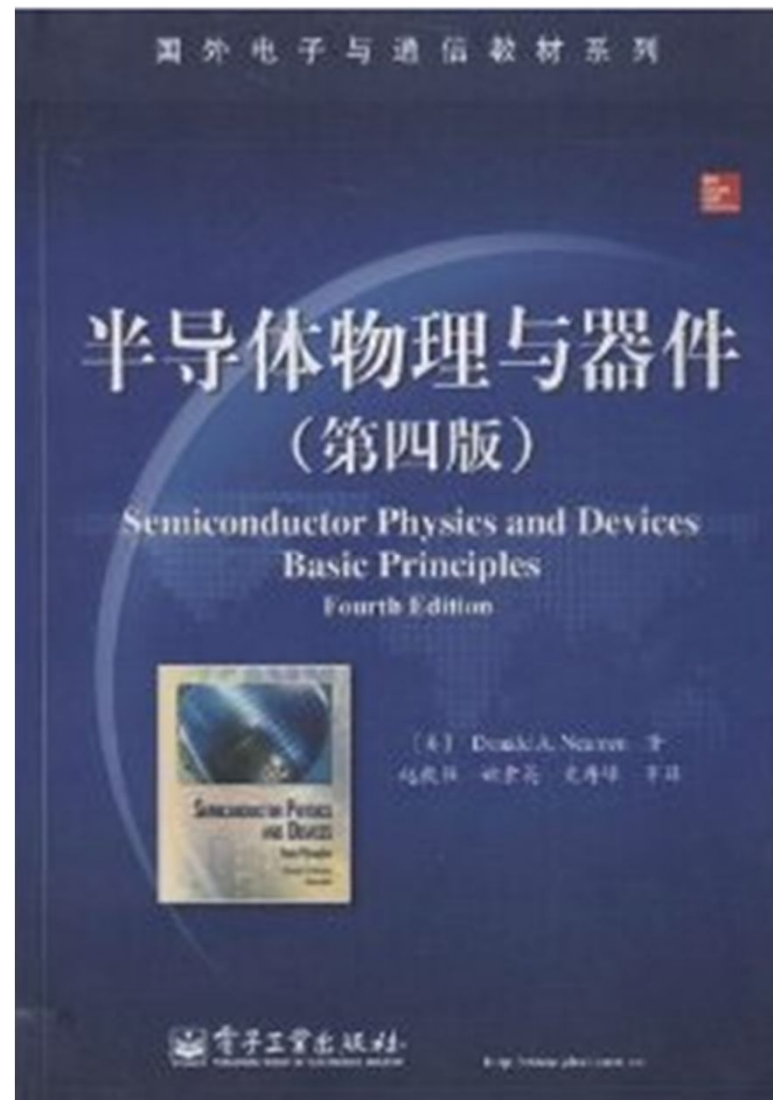
❑ Please change your name
following the format: 19-张三 or
20-李四

❑ Please change your name to let
everyone know who you are.
Students who do not follow the
code will be removed from the
group **without prior notice**.

2月	第1周 春季学期	17 二十	18 雨水	19 廿二	20 廿三	21 廿四	22 廿五	23 廿六
	第2周 春季学期	24 廿七	25 廿八	26 廿九	27 三十	28 二月	1 龙抬头	2 初三
3月	第3周 春季学期	3 初四	4 初五	5 惊蛰	6 初七	7 初八	8 妇女节	9 初十
	第4周 春季学期	10 十一	11 十二	12 植树节	13 十四	14 十五	15 十六	16 十七
	第5周 春季学期	17 十八	18 十九	19 二十	20 春分	21 廿二	22 廿三	23 廿四
	第6周 春季学期	24 廿五	25 廿六	26 廿七	27 廿八	28 廿九	29 三月	30 初二
4月	第7周 春季学期	31 初三	1 初四	2 初五	3 初六	4 清明	5 初八	6 初九
	第8周 期中考试周	7 初十	8 十一	9 十二	10 十三	11 十四	12 十五	13 十六
	第9周 期中考试周	14 十七	15 十八	16 十九	17 二十	18 廿一	19 廿二	20 谷雨
	第10周 春季学期	21 廿四	22 廿五	23 廿六	24 廿七	25 廿八	26 廿九	27 三十
5月	第11周 春季学期	28 四月	29 初二	30 初三	1 劳动节	2 初五	3 初六	4 青年节
	第12周 春季学期	5 立夏	6 初九	7 初十	8 十一	9 十二	10 十三	11 母亲节
	第13周 春季学期	12 十五	13 十六	14 十七	15 十八	16 十九	17 二十	18 廿一
	第14周 春季学期	19 廿二	20 廿三	21 小满	22 廿五	23 廿六	24 廿七	25 廿八
	第15周 春季学期	26 廿九	27 五月	28 初二	29 初三	30 初四	31 端午节	1 儿童节
	第16周 春季学期	2 初七	3 初八	4 初九	5 芒种	6 十一	7 十二	8 十三

Midterm
Exam

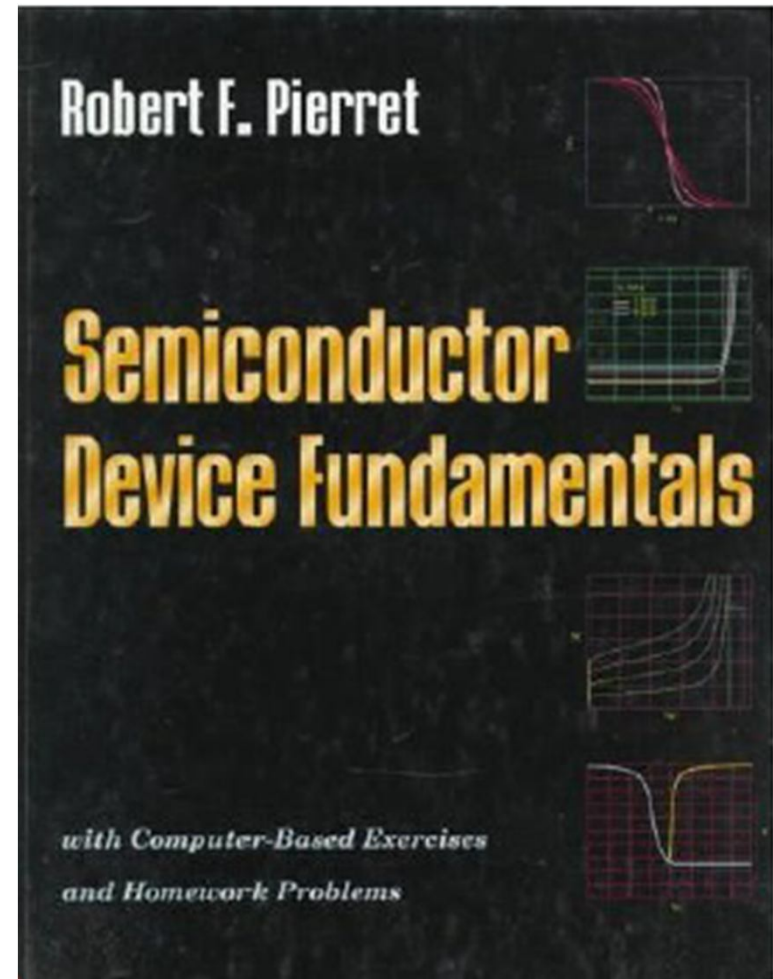
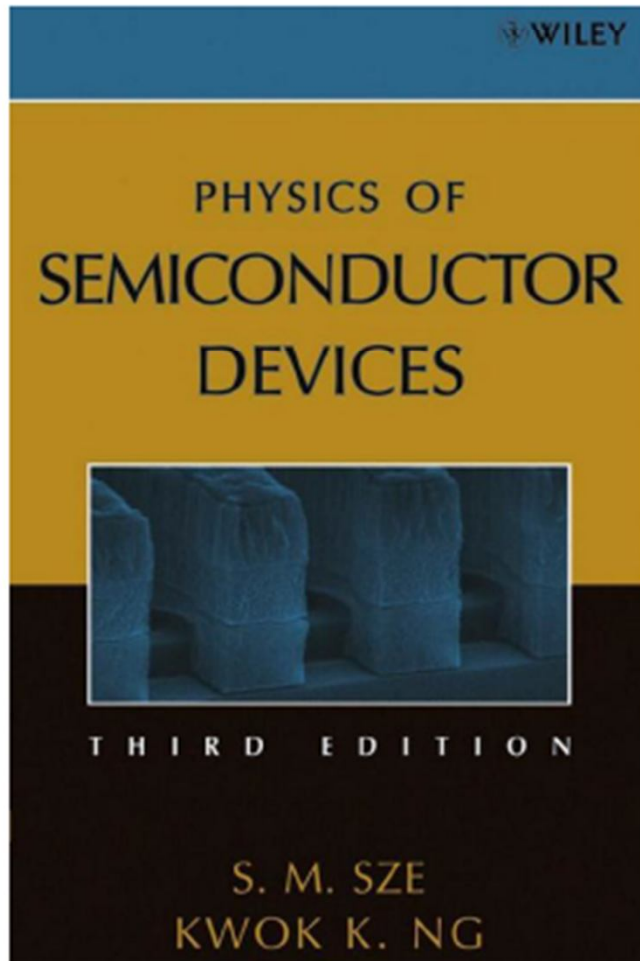
Textbook



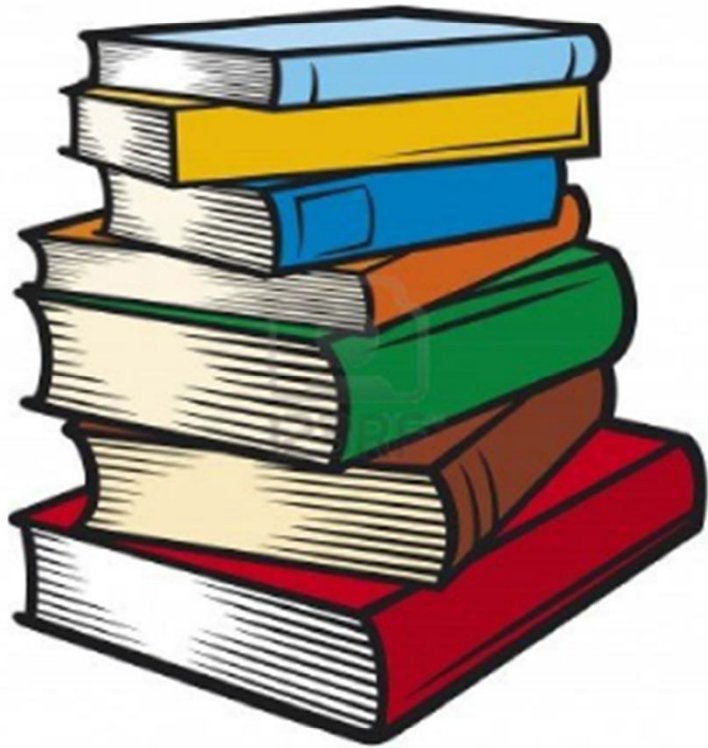


Books : Advanced / Reference

Bible of all devices

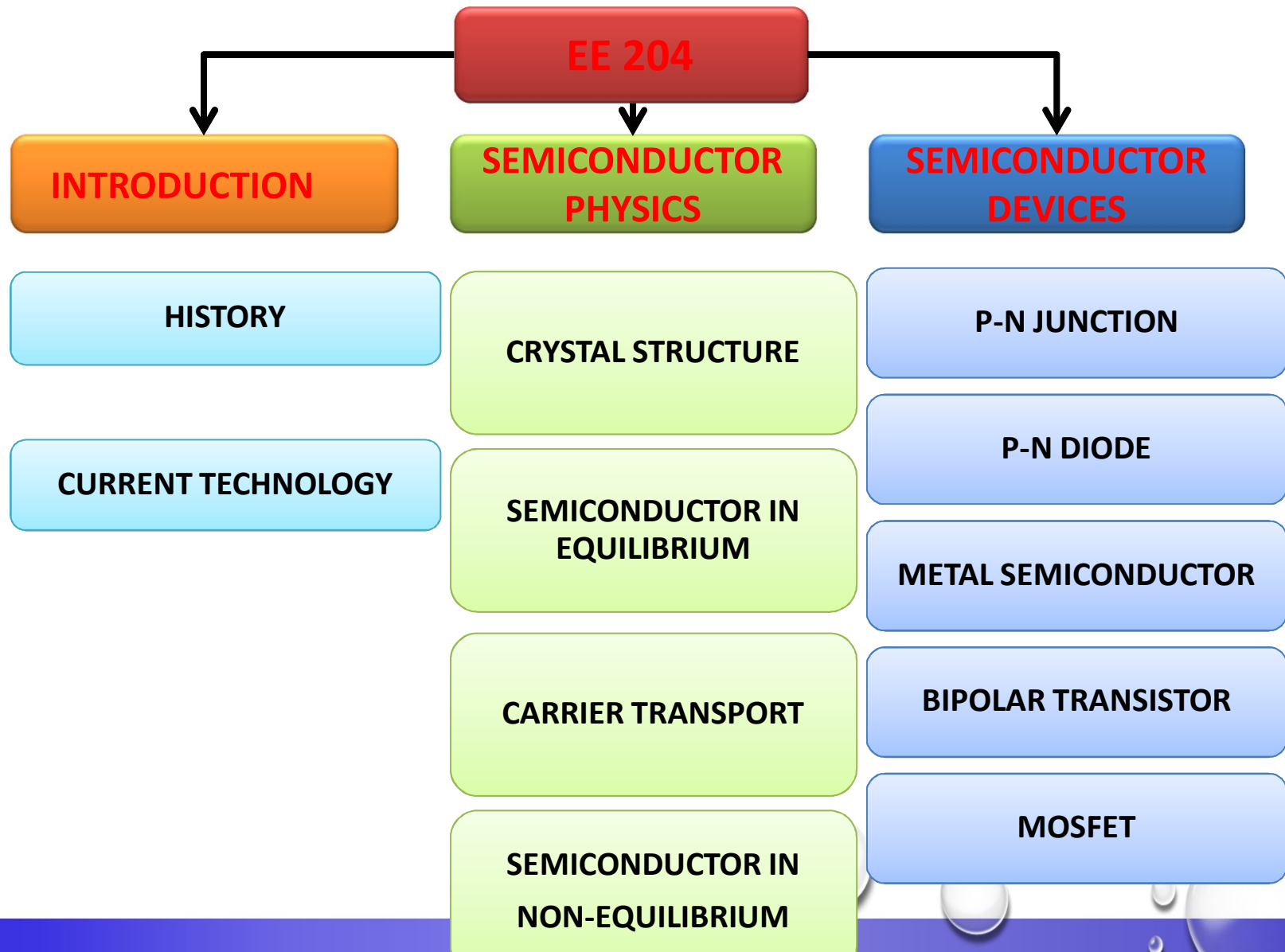


Books



- Browse through the 3 and follow any one you like
- Some of the problems in the problem sets will be taken from the books but you do not need to have the book to solve them
- Follow **at least one** book along with the class note

Major Topics



Tentative schedule

Part 1: Semiconductor Physics

Chapter 1: The Crystal structure of solids

Chapter 3: The energy bands

Chapter 4: Semiconductor in equilibrium

Chapter 5: Carrier transport

Chapter 6: Non-equilibrium excess carrier

Midterm

Part 2: Semiconductor Devices

Chapter 7: pn junction

Chapter 8: pn diode

Chapter 9: MS contact

Chapter 10: MOS capacitor and MOSFET

Chapter 11: Bipolar transistor

Final exam

Lab and project

Simulation (Silvaco) and testing the I-V characteristics of

- 1). p-n diodes
- 2). solar cells
- 3). MOSFETs.

Course grading

Midterm 20%, Final 40%, homework 20%, Lab 20%

Student responsibilities and policies

- ✓ Attending classes;
- ✓ Reading textbook;
- ✓ Completing homework assignments in time - - Late homework will not be accepted;
- ✓ Collaboration policy: Limited homework collaboration. You are welcome to discuss problems and concepts with fellow students verbally, but you should not share written work.

You should take this course if you...

- are interested in semiconductor devices
- would like a career in fab processing, device physics, device design or integration
- would like to be an IC designer who understands devices and can communicate with fab personnel.
- would like to be an IC designer who can understand the limitations of real devices and take advantage of real device performance.
- would like to be an electronics/systems/anything designer who needs to understand real world devices in real world applications.
- like physics
- select MicroE or OptoE as your major

Have to work hard to pass this course

- EE204 would be one of the most difficult courses that you have learned.
- There are many formulas that are hard to understand.
- Also lots of physics.